

Emergency Medicine & First Aid

Critical Protocols, Triage & Life-Saving Interventions

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1. Basic Life Support (BLS)

1.1 Adult BLS Protocol (2020 AHA/ERC Guidelines)

■ ALWAYS call emergency services (or delegate) before commencing CPR. Do not delay resuscitation to seek equipment.

Basic Life Support provides artificial circulation and ventilation to maintain vital organ perfusion until advanced life support and definitive treatment are available. High-quality CPR is the single most impactful intervention for cardiac arrest.

Step-by-Step BLS Sequence

- 1. SAFE: Ensure scene safety for rescuer and victim
- 2. RESPONSIVE: Shout and tap shoulders — 'Are you alright?'
- 3. CALL: If unresponsive — activate emergency medical services (dial 1122/112/911); get AED
- 4. AIRWAY: Head-tilt chin-lift (or jaw thrust if trauma suspected)
- 5. BREATHING: Look, listen, feel for no more than 10 seconds — gasping is NOT normal breathing
- 6. COMPRESSIONS: 30 compressions at 100–120/min, depth 5–6 cm, full chest recoil
- 7. VENTILATION: 2 rescue breaths over 1 second each (30:2 ratio) — if trained
- 8. AED: Power on, attach pads, follow voice prompts; minimise interruptions to compressions
- 9. Continue CPR: Until ROSC, ALS team arrival, or rescuer exhaustion/safety threat

Parameter	Adult	Child (1–8 yrs)	Infant (< 1 yr)
Compression Rate	100–120/min	100–120/min	100–120/min
Compression Depth	5–6 cm	≥5 cm (1/3 AP diameter)	≥4 cm (1/3 AP diameter)
Hand Position	Lower half of sternum	Heel of one hand	2 fingers or 2 thumb-encircling
Compression:Ventilation	30:2 (1 rescuer)	30:2 (1 rescuer); 15:2 (2 rescuer)	30:2 (1 rescuer); 15:2 (2 rescuer)
Shock Energy (AED)	Adult pads	Paediatric pads preferred	Paediatric pads/attenuator

2. Anaphylaxis Management

Anaphylaxis is a severe, life-threatening systemic hypersensitivity reaction characterised by rapid onset. It requires immediate recognition and administration of intramuscular adrenaline.

Diagnostic Criteria

Anaphylaxis is likely if ONE of the following criteria is met within minutes to hours of allergen exposure:

- 1. Sudden onset affecting skin/mucosa PLUS respiratory compromise OR cardiovascular collapse
- 2. Two or more of: skin/mucosal involvement, hypotension, respiratory symptoms, persistent GI symptoms — after allergen exposure
- 3. Hypotension alone after known allergen exposure

Emergency Treatment Protocol

- POSITION: Lie flat with legs elevated (unless dyspnoeic — then sit up; pregnant — left lateral)
- ADRENALINE: Adrenaline (epinephrine) 0.5 mg (1:1000) IM into anterolateral mid-thigh — IMMEDIATELY
- CALL: Activate emergency medical services
- OXYGEN: High-flow via non-rebreather mask 15 L/min
- IV ACCESS: Establish large-bore IV; 500–1000 mL 0.9% NaCl fluid challenge for hypotension
- ADRENALINE: Repeat IM adrenaline every 5 minutes if no improvement
- ADJUNCTS: Chlorphenamine 10 mg IM/IV (antihistamine); Hydrocortisone 200 mg IV — secondary only
- MONITOR: Vital signs continuously; observe minimum 6 hours (12–24 h if severe or biphasic risk)

■ Adrenaline is the ONLY first-line treatment for anaphylaxis. Antihistamines and steroids are adjuncts and must NEVER delay or replace adrenaline administration.

3. Stroke Recognition & Management

3.1 FAST Assessment Tool

Rapid identification of stroke symptoms is critical — for ischaemic stroke, 'time is brain': approximately 1.9 million neurons are lost per minute without treatment.

Letter	Assessment	Positive Finding
F — Face	Ask patient to smile	Facial droop or asymmetry
A — Arms	Ask to raise both arms	One arm drifts down or cannot be raised
S — Speech	Ask to repeat a simple phrase	Slurred, strange, or unable to speak

Letter	Assessment	Positive Finding
T — Time	Call emergency services	ANY positive finding = stroke until proven otherwise

■ **EMERGENCY: Ischaemic stroke — IV thrombolysis (alteplase) window is 4.5 hours from symptom onset. Mechanical thrombectomy window up to 24 hours in selected patients. Do NOT delay — every minute counts.**

3.2 Additional Stroke Symptoms

- Sudden severe headache 'worst of life' (haemorrhagic stroke / SAH)
- Sudden loss of vision in one or both eyes
- Sudden onset dizziness, loss of balance or coordination
- Sudden confusion or difficulty understanding speech (receptive aphasia)
- Sudden numbness of face, arm, or leg — especially unilateral

4. Choking & Airway Obstruction

4.1 Heimlich Manoeuvre (Adult & Child > 1 yr)

Severe airway obstruction (inability to speak, cough, or breathe) requires immediate physical intervention. The Heimlich manoeuvre creates a sudden increase in intrathoracic pressure to expel the foreign body.

- Stand behind the victim; locate navel and lower rib margin
- Deliver 5 firm back blows between shoulder blades with heel of hand
- Make a fist and place thumb-side against abdomen, midline above navel and well below xiphoid
- Grasp fist with other hand; deliver 5 brisk upward abdominal thrusts
- Alternate 5 back blows + 5 abdominal thrusts until obstruction cleared or victim unconscious
- If victim loses consciousness: begin CPR — check airway before each ventilation attempt

■ For infants < 1 year: Use 5 back blows and 5 chest thrusts (NOT abdominal thrusts). For pregnant or obese individuals: Use chest thrusts instead of abdominal thrusts.

5. Triage Categories (START System)

Mass casualty triage uses the START (Simple Triage and Rapid Treatment) protocol to rapidly categorise victims by survival priority when resources are limited.

Colour	Category	Criteria	Action
RED	Immediate	Life-threatening but survivable; abnormal respirations/perfusion/mental status	Treat first
YELLOW	Delayed	Serious injury but stable; can wait without immediate risk to life	Treat second
GREEN	Minor	Walking wounded; minor injuries; can follow instructions	Self-care / delayed
BLACK	Expectant/Deceased	No respirations after airway repositioning, or obviously deceased	Comfort measures only

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